Assignment 04
21020 24

Chapter - 08
Classification : Basic Concepts

Exercise :

8. (1) Major steps of decision tree classification are :

1) First we have to collect and preprocess the training dataset.

2) Then we have to select the best attribute based on the ~~selected~~ measures like information gain.

3) Split the dataset into subsets based on the attribute

4) Finally when all records belong to the same class, or no attribute remain or any depth of tree is defined we can stop.

5) Optionally we can prune the tree and finally use our tree to classify.

8.(2) Pruning is useful as the decision tree may overfit the training data. Tree pruning solves this by removing the least reliable branches.

Seperate set of tuple can be skewed and thus ~~may~~ result can be more reliable. Also our testing set will also decreake.

(8.3) Removing an entire subtree can occur, if we prune before generating rules. So it may remove multiple classification path.

So the first approach, converting the decision tree to rules and then pruning is more flexible. Here rules are independent and seperate.

(8.4) In the worst case scenario, we would have to use as many attributes as possible.

The max depth $\log_2(|D|)$

At each level we have compute $O(n)$

[∵ here, n means one per attribute]

And the total tuple $|D|$

∴ So we will get $O(n \times |D| \times \log|D|)$

(8.6) Naive classification is called naive because it assumes every attribute is independent of each other.

The major idea is:

classes are $c_1, c_2, \ldots, c_m$

And n dimentional vector $X = (x_1, x_2, \ldots, x_n)$

Then we try to maximize $P(c_i|X) = \dfrac{P(X|c_i) \cdot P(c_i)}{P(X)}$

Here $P(X)$ is constant and $P(c_i)$ are avually likely

Then we have to maximize $P(X|c_i)$

For reducing complexity, we consider naive

8.7 (a) As the dataset is generalized the multiple rows are combined into one.

So we can calculate total count like,

$$N = \sum_{i=1}^{n} \text{count}_i$$

$\therefore N = 30 + 40 + 40 + \ldots + 6$

dept	status	age	salary	count
sales	senior	31..35	46k...50k	30

(b) Lets use ID3,

Here,

senior = 52

Junior = 113

total = 165

$$\therefore \text{Intropy}(D) = -\frac{52}{165}\log_2\frac{52}{165} - \frac{113}{165}\log_2\frac{113}{165}$$

$$= 0.8990$$

for dept.

Department	senior	junior	total
sales	30	80	110
systems	8	23	31
marketing	10	4	14
secretery	4	6	10

$\therefore E(\text{sales})_2 = -\frac{30}{110}\log_2\frac{30}{110} - \frac{80}{110}\log_2\frac{80}{110} = 0.845$

$\therefore E(\text{system}) = -\frac{8}{31}\log_2\frac{8}{31} - \frac{23}{31}\log_2\frac{23}{31} = 0.824$

$\therefore E(\text{marketing}) = -\frac{10}{14}\log_2\frac{10}{14} - \frac{4}{14}\log_2\frac{4}{14} = 0.863$

$\therefore E(\text{secretery}) = -\frac{9}{10}\log_2\frac{4}{10} - \frac{6}{10}\log_2\frac{6}{10} = 0.971$

Finally,

$E(\text{dept}) = \frac{110}{165} \cdot 0.84535 + \frac{31}{165} \cdot 0.82381 + \frac{14}{165} \cdot 0.86312$

$+ \frac{10}{165} \cdot 0.97095$

$= 0.85042$

$\therefore \text{Gain}(\text{dept}) = 0.89903 - 0.85042$

$= 0.0486$

For age

Age	Senior	Junior	Total
21-25	0	20	20
26-30	0	49	49
31-35	35	44	79
36-40	10	0	10
41-45	3	0	3
46-50	4	0	4

Here all groups except 31-35 have entropy 0

$$\therefore E(age) = \frac{20}{165}0 + \frac{42}{165}\cdot 0 + \frac{79}{165}\,0.99062 + \frac{10}{165}\cdot 0 + \frac{3}{165}0 + \frac{1}{165}0$$

$$= 0.47930$$

$$Gain(age) = 0.89903 - 0.47930 = 0.4297$$

For salary,

All groups except 46-50k have entropy 0

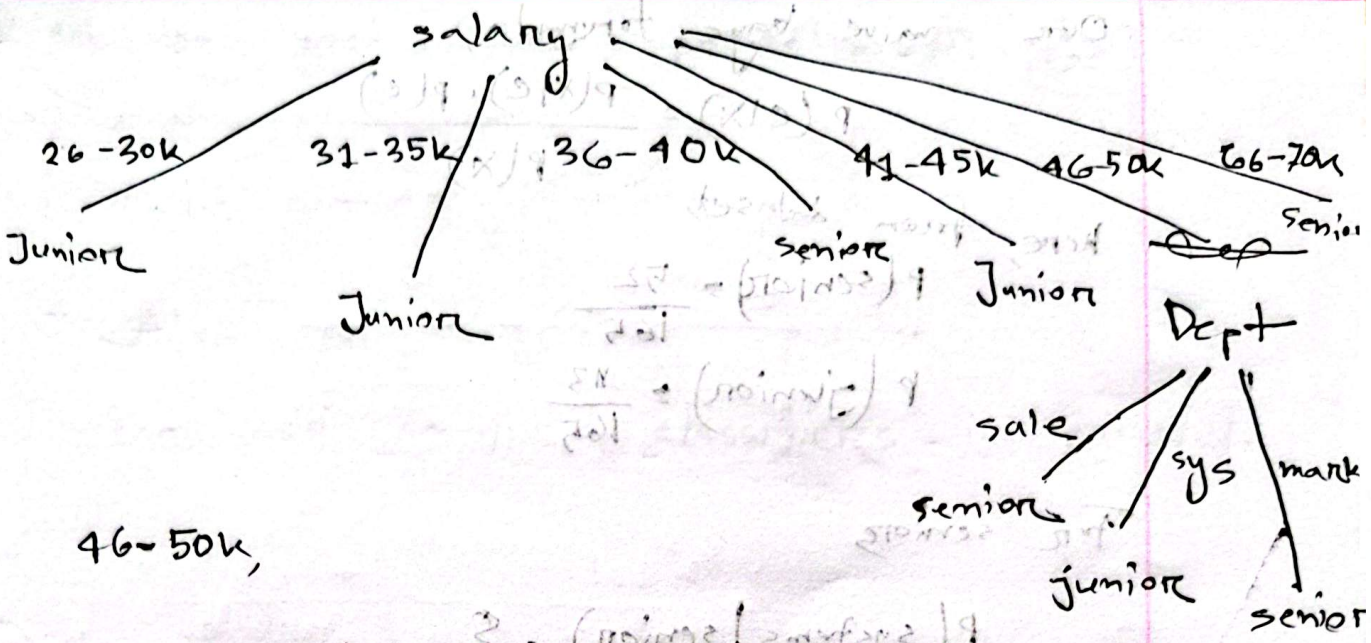
$$\therefore E(salary) = \frac{63}{165}\,0.94602 = 0.36151$$

$$Gain(salary) = 0.89903 - 0.36151$$

$$= 0.5375$$

So here salary has more gain, so it will be selected as the root node.

salary	senior	Junior	total
26-30K	0	46	46
31-35K	0	40	40
36-40K	4	0	4
41-45K	0	4	4
46-50K	40	23	63
66-70K	8	0	8



For 46-50k,

dept	status	age	count
sales	senior	31...35	30
systems	junior	21...25	20
"	"	26...30	3
marketing	senior	36...40	10

So, we split simply based on department here.

ⓒ Our new tuple is,

dept = system

salary ~~status~~ = 46...50k

age = 26...30

Our naive bays formula,

$$P(c|x) = \frac{P(x|c) \cdot P(c)}{P(x)}$$

Here, from dataset,

$$P(\text{senior}) = \frac{52}{165}$$

$$P(\text{junior}) = \frac{113}{165}$$

For seniors

$$P(\text{systems} \mid \text{senior}) = \frac{8}{52}$$

$$P(\text{Age}_{26\text{-}30} \mid \text{senior}) = \frac{0}{52}$$

$$P(\text{salary}_{40\text{-}50} \mid \text{senior}) = \frac{40}{52}$$

$$\therefore P(\text{senior}|x) = \frac{52}{165} \times \frac{8}{52} \times \frac{0}{52} \times \frac{40}{52} = 0$$

Again,

$$P(\text{systems} \mid \text{junior}) = \frac{23}{113}$$

$$P(\text{Age}_{26\text{-}30} \mid \text{junior}) = \frac{49}{113}$$

$$P(\text{salary}_{40\text{-}50} \mid \text{junior}) = \frac{23}{113}$$

$$\therefore P(\text{junior}|x) = \frac{113}{165} \times \frac{23}{113} \times \frac{49}{113} \times \frac{23}{113}$$

So it has more probability

(8.9) We can use naive approach. As new streams will come, we count and update the table accordingly.

Also for improving performance we can add more classifiers in parallel.

(8.10) Here,

$$\text{sensitivity} = \frac{TP}{P}$$

$$\text{specificity} = \frac{TN}{N}$$

And $$\text{accuracy} = \frac{TP + TN}{P+N}$$

$$= \frac{TP}{P+N} + \frac{TN}{P+N}$$

$$= \frac{TP}{P} \times \frac{P}{(P+N)} + \frac{TN}{N} \times \frac{N}{(P+N)}$$

So, we can say,

accuracy is a function of sensitivity and specificity.

8.11 From the above description,

if F is the measure of harmonic mean of precision and recall,

$$F = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}}$$

$$= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

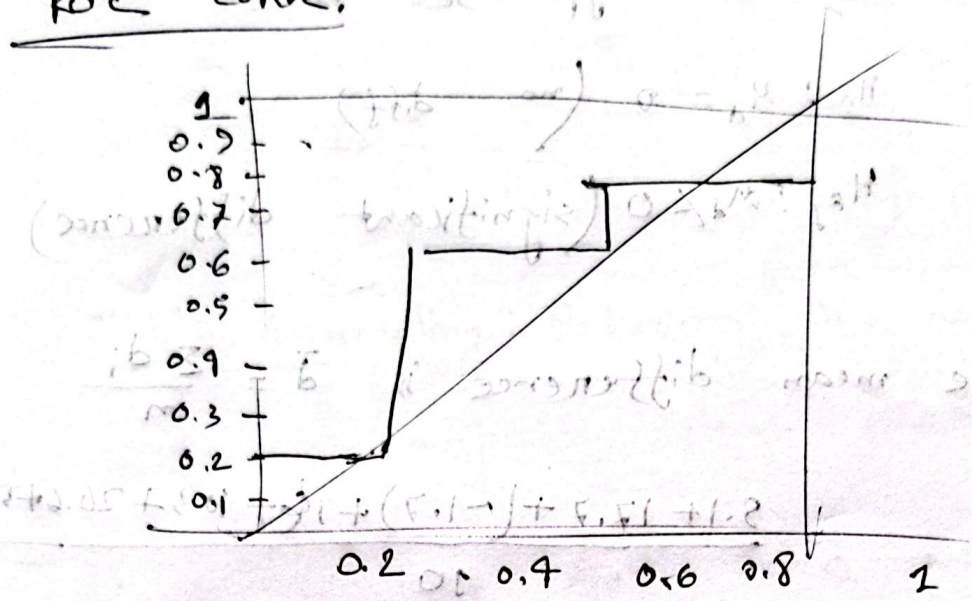
And

$$F_\beta = \frac{(1+\beta^2) \cdot TP}{(1+\beta^2) \times TP + (\beta^2 \times FN) + FP}$$

8.12

Tuple	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	p	0.95	1	0	5	4	0.2	0
2	n	0.85	1	1	4	4	0.2	0.2
3	p	0.78	2	1	4	3	0.4	0.2
4	p	0.66	3	1	4	2	0.6	0.2
5	n	0.60	3	2	3	2	0.6	0.4
6	p	0.55	4	2	3	1	0.8	0.4
7	n	0.53	4	3	2	1	0.8	0.6
8	n	0.52	4	4	1	1	0.8	0.8
9	n	0.51	4	5	0	1	0.8	1
10	p	0.4	5	5	0	0	1	0

ROC curve:



8.13 A data object may belong to more than one class at a time. However the data object will tend to skew in one direction more than other.

For example, we can use classification accuracy, other heuristic to choose the classifier. We can also consider speed, robustness, scalability and interpretability.

8.14 Here, our hypotheses is

$H_0 : \mu_d = 0$ (no diff)

$H_{0_1} : \mu_d \neq 0$ (significant difference)

So the mean difference is $\bar{d} = \frac{\sum d_i}{n}$

$$= \frac{8.1 + 17.7 + (-1.7) + 16 + 10.3 + 20.6 + 5.6 + 6.6 + 5.3}{10}$$

$$= 6.45$$

Standard deviation $S_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}}$

$$= 8.7001$$

For t-test $t = \frac{\bar{d}}{\frac{S_d}{\sqrt{n}}} = \frac{6.45}{\frac{8.7001}{\sqrt{10}}} = 2.3444$

Here,

Significance level $\alpha = 0.01$

degrees of freedom = 9

So the critical value = 3.25

As $2.3444 < 3.25$.

we can't reject the hypothesis

So they are not significantly different

(8.16) Methods for addressing class imbalance problem:

1. Oversampling: Oversampling or increasing the number of minority class
2. Undersampling: Reducing the number of majority class examples.
3. Cost-sensitive: Add more misclassification cost to the minority class.
4. Ensemble learning: Combining multiple classifiers to improve minority class detection.
5. Threshold: Also we can adjust threshold to detect more minority class

For detecting fraud in bank we can use oversampling to increase the fraud examples.